Red Light and Speed Camera Deployment and Traffic Crash Outcomes in Chicago

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1. Research Question

Traffic accidents pose serious risks to public safety and generate substantial social and economic costs. Many cities have adopted automated enforcement tools such as **red light cameras and speed cameras** to improve traffic safety and encourage compliance with traffic laws. Chicago has deployed a large number of these cameras as part of its traffic safety strategy.
This project examines the relationship between automated traffic enforcement and traffic crash outcomes in Chicago. Specifically, we focus on two main research questions:

1. Are traffic cameras placed in areas with higher crash risk?
2. Do traffic cameras reduce crash frequency or crash severity?

2. Data and Methodology

All datasets used in this project are sourced from the Chicago Data Portal.

We combine four datasets from the Chicago Data Portal: Chicago neighborhood boundaries, traffic crash records, speed camera locations, and red light camera locations.

Before conducting the analysis, we cleaned the datasets using Python and pandas. This included converting crash dates into datetime format, standardizing time variables, filtering observations with missing coordinates, and preparing geographic coordinates for spatial analysis. Because earlier years contain incomplete monthly records and substantially fewer crash observations, we restrict our analysis to 2018 onward, when the data coverage becomes stable.

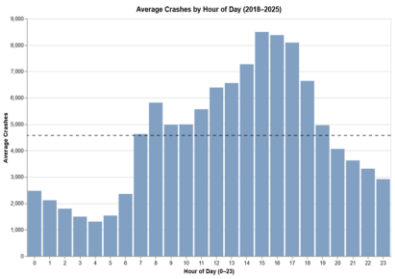
Descriptive Analysis: We first examine general crash patterns in Chicago, including temporal trends, crash severity distribution, and the relationship between speed limits and fatal crash risk. This helps us to better understand the overall characteristics of the data.

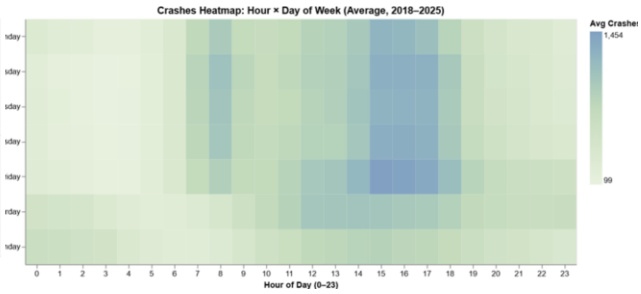
Spatial Analysis: We then visualize the spatial distribution of crashes and compare them with the locations of speed cameras and red light cameras.

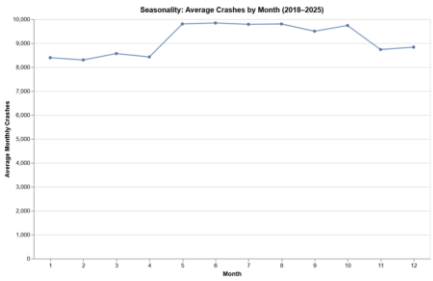
Event Study Design: To evaluate whether traffic cameras are associated with changes in crash outcomes, we implement an event-study design that aligns crash outcomes relative to each camera's installation date. We define treatment exposure using a 200-meter buffer around each camera. This approach allows us to observe how crash patterns evolve in the months before and after treatment.

3. Results

Temporal Patterns of Traffic Crashes



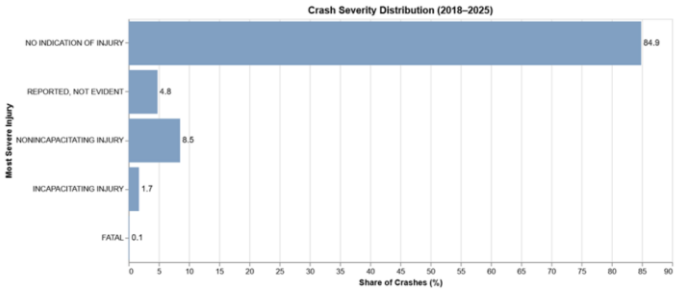


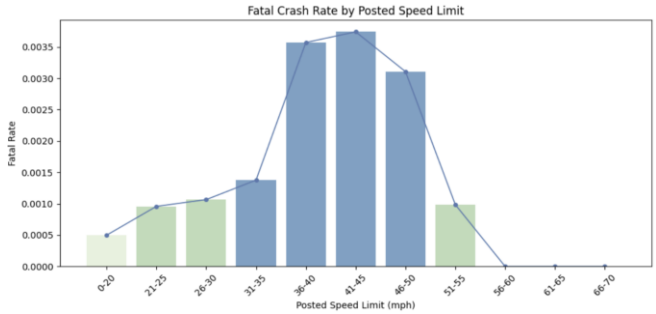


Crash frequency varies substantially across hours, days, and months. Crashes tend to peak during afternoon rush hours and are most common on weekday afternoons, especially Fridays. Crash frequency also increases during late

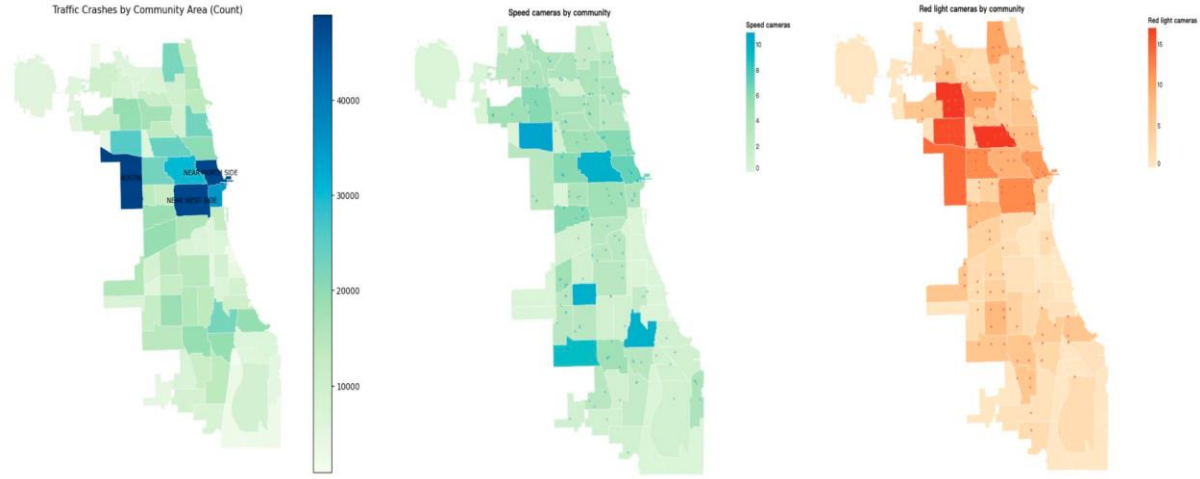
spring and summer, likely reflecting travel activity.

Crash Severity and Speed Limits

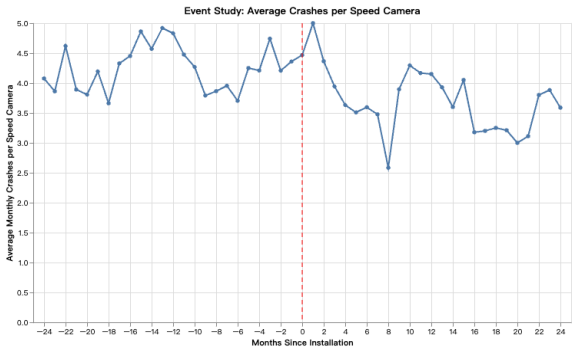


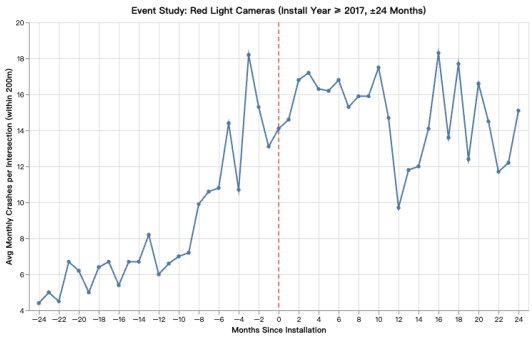


Most crashes involve no reported injury, while severe injuries and fatalities account for only a small share of incidents. We find a clear relationship between posted speed limits and fatal crash risk: the fatal crash rate increases as speed limits rise and peaks around 40–50 mph. At very high speed limits, the fatal rate declines slightly, possibly because these roads are typically highways with fewer intersections and more controlled traffic flow. Overall, these results highlight the importance of speed regulation in reducing crash severity.



We compare the geographic distribution of traffic crashes and camera deployments across Chicago community areas. The crash heatmap shows that traffic crashes are unevenly distributed across the city, with clear clustering in high-activity areas. We also map the locations of speed cameras and red light cameras. Camera placement partially overlaps with areas that experience high crash frequencies. These patterns motivate our event-study analysis.





Speed Cameras

Since our crash dataset begins in 2013, cameras installed earlier lack a sufficiently long observable pre-

treatment period. As a result, we restrict our analysis to cameras installed during the recent expansion beginning in 2023. The event-study results show that crash levels remain relatively stable during the pre-treatment period, suggesting that there is no evidence of anticipatory behavior or violations of the parallel pre-trend assumption. After installation, the plot shows an estimated reduction of roughly 15-25 percent in crash frequency. Overall, these findings suggest that speed cameras are associated with a meaningful reduction in local traffic crashes.

Red Light Cameras

We apply the same framework for red light cameras. The only difference is that we have to restrict our analysis to intersections where red light cameras were installed after 2017 due to data limitations. The results differ notably from those for speed cameras. In the pre-treatment period, crash counts show a clear upward trend, which may indicate that intersections experiencing rising crash risk are more likely to receive red light cameras. At the time of installation, there is no immediate decline in crash counts, but crashes appear to stabilize in the post-treatment period rather than continuing to increase. However, the small data sample may contribute to unstable estimates, so we can not make a credible causal claim.

4. Streamlit Dashboard

In addition to static visualizations, we developed an **interactive Streamlit dashboard** that allows users to explore crash data and camera deployment patterns dynamically. The dashboard can be accessed at: https://final-project-chicago-crashes-g9srdmczqbxunyfhgnttyg.streamlit.app/

Our Streamlit app visualizes how traffic crashes are distributed across Chicago community areas and how that distribution changes by hour of day and crash type, while also overlaying the locations of speed cameras and red-light cameras.

This directly supports our research question by letting us assess whether automated enforcement infrastructure is concentrated in high-risk areas and whether the spatial pattern of camera placement aligns with where and when crashes are most frequent. Also, Users can interact with the dashboard to explore crash patterns across locations and daytime periods. This interactive tool helps policymakers identify high-risk areas and evaluate where additional traffic enforcement measures may be most effective.

5. Limitations and Challenges

This analysis has several limitations. First, the crash dataset begins in 2013 and appears more consistent only after 2018, which limits our ability to analyze earlier camera deployment. As a result, many red light cameras installed before 2010 cannot be included in the event-study analysis, leading to a relatively small sample size. Second, although the event-study framework helps compare crash patterns before and after installation, it does not fully control for other factors that may influence crashes, such as changes in traffic volume, road conditions, or other safety policies. Therefore, the results should be interpreted as suggestive associations rather than definitive causal effects.

6. Policy Implications

Our findings provide several insights for traffic safety policy in Chicago. First, the results suggest that speed cameras are associated with a reduction in crash frequency in areas where they are installed. This indicates that automated speed enforcement may be an effective tool for improving road safety.

Second, the spatial analysis shows that traffic crashes are unevenly distributed across the city, with clear clustering in high-activity areas. This suggests that traffic safety interventions, including camera deployment, could be more effective if they are strategically targeted toward high-risk locations and peak traffic periods.

Finally, the results highlight the importance of using data-driven approaches to guide traffic safety policies. By combining crash data with information on traffic enforcement infrastructure, policymakers can better identify where additional cameras or other safety interventions may generate the greatest safety benefits.